



SFAS IoT Sensor Setup Guide

Complete Step-by-Step Configuration Manual

ESP8266 NodeMCU · DHT22 Temperature & Humidity · FC-28 Soil Moisture
Smart Farming Advisory System — Nyagatare District, Rwanda

Hardware	ESP8266 NodeMCU v1.0 or v3 + DHT22 + FC-28 Soil Sensor
Firmware file	sfas_sensor.ino (upload with Arduino IDE)
Server file	modules/IoT/api/iotApi.php on your XAMPP
DB table	iot_readings + iot_devices (run iot_extension.sql)
Dashboard	modules/IoT/views/iot-dashboard.php
Route	/api/iot and /admin/iot-dashboard in index.php

1. What You Have & What Each Does

You have three hardware components. Here is exactly what each one does:

Component	Measures	How It Connects
ESP8266 NodeMCU	— (it is the brain)	USB cable to your laptop. Runs the firmware and sends data over WiFi.
DHT22 Sensor	Temperature (°C) + Humidity (%)	4 pins: VCC → 3.3V, DATA → D4, (skip NC pin), GND → GND
FC-28 Soil Moisture	Soil moisture 0–100%	4 pins: VCC → 3.3V, GND → GND, AOUT → A0, DOUT → not used

□ Why these three sensors?

Temperature and humidity from the DHT22 let the system detect conditions that favor fungal diseases (high humidity >85%) and heat stress (temperature >32°C). The soil moisture sensor tells the system when the soil is dry enough that irrigation is needed — the AI assistant uses this context when giving farming advice.

2. Wiring Diagram — Connect Everything

Complete connection table. Do this before installing any software. Use jumper wires.

2.1 DHT22 Sensor Connections

DHT22 Pin	Label on sensor	Connect to NodeMCU	Note
Pin 1 (left)	VCC or +	3.3V (NOT 5V)	Powers the sensor — must be 3.3V not 5V
Pin 2	DATA or OUT	D4 (GPIO2)	Signal wire — put 10kΩ resistor between here and VCC
Pin 3	NC	Leave unconnected	Not connected — skip this pin
Pin 4 (right)	GND or –	GND	Ground

❏ Important: 10kΩ Pull-up Resistor Required

The DHT22 DATA pin needs a 10kΩ resistor between it and VCC (3.3V). Without it you will get "nan" readings. Connect one leg to the DATA wire going to D4, other leg to the 3.3V wire. This is a small brown/orange component. If you do not have one, try the sensor anyway — some modules have it built in on the board.

2.2 Soil Moisture Sensor (FC-28) Connections

FC-28 Pin	Label	Connect to NodeMCU	Note
Pin 1	VCC	3.3V	Power — use 3.3V for NodeMCU
Pin 2	GND	GND	Ground
Pin 3	AOUT	A0	IMPORTANT: This is the one you need. Analog output.
Pin 4	DOUT	Leave unconnected	Digital output — not used in SFAS

❏ NodeMCU A0 Pin Limitation

The NodeMCU ESP8266 has only ONE analog input pin — A0. This means you can only connect ONE analog sensor. The soil moisture AOUT goes here. If you want to add a second analog sensor later you need a multiplexer chip (CD4051 or similar).

2.3 Quick Reference — All Connections

NodeMCU Pin	Connects To	What It Is
3.3V	DHT22 VCC + FC-28 VCC	Power supply for both sensors

NodeMCU Pin	Connects To	What It Is
GND	DHT22 GND + FC-28 GND	Ground for both sensors (share one GND pin)
D4	DHT22 DATA	Temperature + Humidity signal
A0	FC-28 AOUT	Soil moisture analog signal
USB (micro-USB)	Your laptop USB port	Powers the NodeMCU and uploads firmware

3. Arduino IDE Setup (Do This Once)

Install the Arduino IDE and configure it for the ESP8266. Do this once — you never need to do it again.

1	Download Arduino IDE Go to https://www.arduino.cc/en/software — Download the Windows installer. Install it with default options.
2	Add ESP8266 Board Support Open Arduino IDE → File → Preferences. In the "Additional Board Manager URLs" box, paste exactly:
3	Install ESP8266 Package Tools → Board → Boards Manager → search for "ESP8266" → click Install on the "esp8266 by ESP8266 Community" package. Wait for download to finish (~50MB).
4	Select the Right Board Tools → Board → ESP8266 Boards → select "NodeMCU 1.0 (ESP-12E Module)"
5	Select Upload Speed Tools → Upload Speed → 115200
6	Install DHT22 Library Tools → Manage Libraries → search "DHT sensor library" → Install "DHT sensor library by Adafruit" → also install "Adafruit Unified Sensor" when asked

ESP8266 board manager URL — copy this exactly:

```
https://arduino.esp8266.com/stable/package\_esp8266com\_index.json
```

☐ How to know it worked

After installing the ESP8266 package, go to Tools → Board — you should see a new section called "ESP8266 Boards" with "NodeMCU 1.0 (ESP-12E Module)" in the list.

4. Calibrate the Soil Moisture Sensor (Do This First)

Every FC-28 sensor gives slightly different numbers. You need to find YOUR sensor's dry and wet values before the main code will give accurate percentage readings.

4.1 Upload the Calibration Sketch

1. Connect the NodeMCU to your laptop via USB
2. Open Arduino IDE → File → Open → navigate to `sfas_calibrate.ino`
3. Verify the board is set to NodeMCU 1.0 (Tools → Board)
4. Click the Upload button (right arrow →) and wait for "Done uploading"
5. Open Serial Monitor: Tools → Serial Monitor
6. Set baud rate to 115200 in the bottom-right dropdown of Serial Monitor

4.2 Find Your DRY Value

7. With the soil sensor connected to A0, hold the sensor probe in DRY AIR — not touching soil, water, or anything
8. Watch the Serial Monitor for about 10 seconds
9. The numbers showing are your DRY readings — write down the highest stable number

Example: if you see 950, 948, 952, 949 → your SOIL_DRY_VALUE = 950

4.3 Find Your WET Value

10. Fill a glass or bowl with plain water
11. Dip the sensor probe (the two metal rods, NOT the electronics board) into the water
12. Watch the Serial Monitor for about 10 seconds
13. Write down the lowest stable number you see

Example: if you see 290, 285, 292, 288 → your SOIL_WET_VALUE = 285

4.4 Update the Main Sketch

Open `sfas_sensor.ino` and find these two lines near the top and replace with YOUR values:

```
const int SOIL_DRY_VALUE = 950;    // ← replace 950 with YOUR dry value
const int SOIL_WET_VALUE = 285;    // ← replace 285 with YOUR wet value
```

□ Typical FC-28 values

Most FC-28 sensors read around 900–1023 when dry and 200–400 when wet. If your Serial Monitor shows 0 for soil, check that AOUT is connected to A0 (not DOUT). If it shows a constant 1023, the sensor may need 5V — try connecting VCC to the VU or VIN pin on NodeMCU (which outputs 5V from USB).

5. Configure the Main Sketch (sfas_sensor.ino)

Open sfas_sensor.ino in Arduino IDE. Find Section ① at the top — these are the ONLY lines you need to edit.

5.1 Find Your Computer's IP Address

The ESP8266 needs to know your XAMPP computer's IP address on the local WiFi network:

- On Windows: press Windows key + R → type cmd → press Enter → type: ipconfig → press Enter
- Look for "Wireless LAN adapter Wi-Fi" section → "IPv4 Address" → e.g. 192.168.1.105
- On the same WiFi: the ESP8266 and your XAMPP computer MUST be on the same WiFi network

❏ Same WiFi network required

Your laptop (running XAMPP) and the NodeMCU MUST be on the same WiFi network. If your laptop is on university WiFi and your phone hotspot is different, it will not work. Use your phone hotspot for both, or connect laptop and ESP to the same home/office WiFi.

5.2 Edit These 5 Lines Only

```
// ===== EDIT THESE 5 LINES =====

const char* WIFI_SSID = "YOUR_WIFI_NAME"; // ← your WiFi name
const char* WIFI_PASS = "YOUR_WIFI_PASSWORD"; // ← your WiFi password

const char* SERVER_IP = "192.168.1.105"; // ← YOUR PC IP from ipconfig

const int SOIL_DRY_VALUE = 950; // ← from calibration Step 4.2
const int SOIL_WET_VALUE = 285; // ← from calibration Step 4.3

// Leave everything else unchanged
```

5.3 Also update your .env file on XAMPP

Open the .env file in your SFAS project root and add this line if it is not there:

```
IOT_SECRET=sfas_iot_secret_2026

# This must match the IOT_SECRET in sfas_sensor.ino exactly
```

5.4 Add new routes to index.php

Open your SFAS index.php router and add these two routes inside the \$routes array:

```
'/api/iot' => 'modules/IoT/api/iotApi.php',
'/admin/iot-dashboard' => 'modules/IoT/views/iot-dashboard.php',
```

5.5 Run the SQL extension

14. Open phpMyAdmin → select sfas_db
15. Click Import tab → choose iot_extension.sql → click Go
16. You should see "Import has been successfully finished"
17. Under Tables you will now see iot_readings and iot_devices

6. Upload the Firmware to the ESP8266

- 1 Connect NodeMCU to laptop**
Use a micro-USB cable. Windows should make a sound when it connects. Check Device Manager — you should see a new COM port (e.g. COM5 or COM6).
- 2 Select the COM port**
Arduino IDE → Tools → Port → select the COM port that appeared (e.g. COM5). If you see multiple, unplug the NodeMCU and see which one disappears — that is the right one.
- 3 Open sfas_sensor.ino**
File → Open → select sfas_sensor.ino. Make sure you edited the 5 settings in Section 5.2 above.
- 4 Verify first (optional)**
Click the checkmark ✓ button (Verify). This compiles the code without uploading. If you see errors, read the error message — most common is a missing library.
- 5 Upload**
Click the right-arrow → button (Upload). The IDE will compile then upload. You will see progress dots in the status bar. NodeMCU LED will flicker.
- 6 Open Serial Monitor**
Tools → Serial Monitor. Set baud rate to 115200. You should see the boot messages within 5 seconds.

□ What success looks like in Serial Monitor

You should see: "[WiFi] Connecting to: YourWiFiName" "[WiFi] ✓ Connected!" "[WiFi] IP Address: 192.168.1.108" "[SENSOR] Reading sensors..." " Temperature : 24.5 °C" " Humidity : 68.2 %" " Soil moisture: 42 %" "[HTTP] Response code: 200" "[HTTP] ✓ Data sent successfully!"

7. Verify Everything is Working

7.1 Check the Serial Monitor Output

What you see	What it means
[WiFi] ✓ Connected!	WiFi working — good
Temperature: 24.5°C	DHT22 working — good
Soil moisture: 42%	Soil sensor working — good
Response code: 200	SFAS received data — everything working!
[WiFi] TIMEOUT	Wrong WiFi name or password — check WIFI_SSID and WIFI_PASS
Temperature: nan	DHT22 not connected right — check wiring, add 10kΩ resistor
Soil raw: 0	Soil sensor AOUT not on A0 — check wiring
Response code: 403	Wrong IOT_SECRET — check .env matches sfas_sensor.ino
Connection FAILED	Wrong SERVER_IP — run ipconfig again and check

7.2 Verify in phpMyAdmin

18. Open phpMyAdmin → sfas_db → iot_readings table
19. Click Browse — you should see rows appearing every 30 seconds
20. If rows are appearing, the sensor is working end-to-end

7.3 Verify on SFAS Dashboard

21. Log in to SFAS at <http://localhost/sfas>
22. Click IoT Sensors in the navigation
23. You should see the live green status dot and current readings
24. The temperature and humidity numbers should match what Serial Monitor shows

8. Troubleshooting

8.1 DHT22 Problems

Problem: Reads "nan" (not a number)

- Check VCC is connected to 3.3V (not 5V)
- Check DATA wire is on D4
- Add a 10kΩ pull-up resistor between DATA pin and VCC pin
- Try a fresh jumper wire on the DATA connection
- Wait 3 seconds after power-on before first read — DHT22 needs warm-up time

Problem: Reads same value every time (stuck reading)

- Unplug and replug the USB cable to restart the NodeMCU
- Check that your jumper wires are firmly in the NodeMCU holes

8.2 Soil Moisture Sensor Problems

Problem: Always reads 0%

- Check AOUT (not DOUT) is connected to A0
- Check VCC is connected — try 5V (VU or VIN pin) if 3.3V gives 0

Problem: Reads 100% even in dry soil

- Your SOIL_DRY_VALUE is too low — redo the calibration with the sensor in dry air
- Check that GND is properly connected

Problem: Percentage is negative or above 100%

- Your calibration values are swapped or wrong — redo calibration
- The code uses constrain() to clamp 0–100 so this should not show in the app

8.3 WiFi / HTTP Problems

Problem: WiFi keeps timing out

- Check WIFI_SSID — WiFi names are case-sensitive ("MyWiFi" ≠ "mywifi")
- Check WIFI_PASS — look for spaces at the start/end, copy carefully
- Make sure your phone hotspot or router is on 2.4GHz — ESP8266 does NOT support 5GHz WiFi
- Move the NodeMCU closer to the router if signal is weak

Problem: HTTP response code 403

- IOT_SECRET in sfas_sensor.ino does not match IOT_SECRET in .env
- Both must be exactly: sfas_iot_secret_2026 (or whatever you set)
- Restart Apache after changing .env

Problem: Connection FAILED — cannot reach server

- Run ipconfig on your laptop and confirm the IP address
- Laptop and NodeMCU must be on the SAME WiFi network
- Apache must be running in XAMPP — check XAMPP Control Panel

- Try opening `http://192.168.1.X/sfas` in your phone browser (same network) — if it opens, the address is correct
- Disable Windows Firewall temporarily to test — if it works, add a firewall rule for port 80

Problem: HTTP code 200 but no data in database

- Run `iot_extension.sql` in phpMyAdmin — the `iot_readings` table may not exist
- Check `/api/iot` route exists in `index.php`
- Check `iotApi.php` path is correct in `index.php` route

□ Quick network test from your phone

While the NodeMCU is trying to connect, open your phone browser (connected to the same WiFi) and go to `http://192.168.1.X/sfas/api/iot?action=latest` (use your PC's IP). If you see a JSON response, your server is reachable. If the page does not open, the IP is wrong or Apache is not running.

9. What to Say at Your Defense

When the panel asks about IoT, use this structured answer:

Panel Question: "How does your IoT integration work?"

Your Answer:

"SFAS uses an ESP8266 NodeMCU microcontroller connected to two sensors in the field: a DHT22 for air temperature and humidity, and an FC-28 capacitive soil moisture sensor. The node reads all sensors every 30 seconds and sends the data as a JSON HTTP POST to a dedicated REST API endpoint — `/api/iot?action=ingest` — in the SFAS backend. The data is stored in the `iot_readings` table in our MySQL database and displayed in real-time on the IoT Dashboard page, which auto-refreshes every 30 seconds."

Panel Question: "Why IoT and not just weather API data?"

Your Answer:

"Open-Meteo gives regional weather data for the whole Nyagatare District. But microclimate conditions on a specific farm can be significantly different — a farm in a valley is cooler and more humid than one on a hillside 2km away. The IoT sensor captures hyperlocal data that the regional API cannot provide. The SFAS IoT Dashboard shows both readings side by side so farmers and agronomists can see the difference. Additionally, the soil moisture sensor provides data that no weather API offers — it tells us the actual water content in the soil, which is critical for irrigation decisions."

Panel Question: "What happens if the sensor goes offline?"

Your Answer:

"The system handles this gracefully. The dashboard checks the timestamp of the last reading — if it is more than 2 minutes old, the status dot turns red and shows Offline. The rest of SFAS continues to function normally using the Open-Meteo weather API. The database stores all historical readings so when the sensor reconnects, the history chart picks up from where it left off. The firmware on the ESP8266 also has automatic WiFi reconnection logic and restarts itself if it fails 10 consecutive times."

10. Hardware Shopping List

Everything you need to buy in Kigali. All available at electronics shops in Nyarugenge (near UTC/Kigali City Tower area) or on Jumia Rwanda.

Item	Where to buy	Approx cost (RWF)	Notes
ESP8266 NodeMCU v1.0 or v3	Kigali electronics / Jumia	8,000–12,000	Any version works. v3 is newer.
DHT22 temperature & humidity sensor	Kigali electronics / Jumia	2,000–4,000	Sometimes sold as AM2302.
FC-28 soil moisture sensor module	Kigali electronics / Jumia	1,500–3,000	Buy the module (with board), not bare probes.
Jumper wires (male-to-male + male-to-female)	Kigali electronics	500–1,000	Pack of 40 is more than enough.
10kΩ resistor (for DHT22 pull-up)	Kigali electronics	100–200	Buy a pack of 10 — very cheap.
Micro-USB cable	Any phone shop	1,000–2,000	For uploading firmware and power.
Breadboard (optional, makes wiring easier)	Kigali electronics	1,500–2,500	Strongly recommended for demo.
TOTAL		~15,000–25,000 RWF	

□ Buying tip

If you cannot find the parts in Kigali in time, you can also order from Jumia Kenya ([jumia.co.ke](https://www.jumia.co.ke)) with 2–3 day delivery. Search: "ESP8266 NodeMCU" and "DHT22 module" and "soil moisture sensor module FC-28". Make sure to order the MODULE versions which include the PCB board, not just bare components.